This means that, if we use curves instead of segments, computing the edge crossing at each iteration of an optimization algorithm is going to be much more expensive.

Moreover, with quadratic curves we already have 4 possible intersections for each pair of edges, and for cubic curves it’s obviously even worse: we have more parameters to optimize, and each change can make a greater difference.

That’s why we are focusing on straight-line drawings: one should really double check the requirements and discuss the benefits and costs, before deciding to embark in the more challenging enterprise of curve-line embeddings.

